

DPS Hand: A Dual-Parallelogram and Slot-Guided Gripper for Adaptive Grasping and Environmental Scooping

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Abstract—Traditional robotic hands with parallel or coupled linkages often suffer from curved fingertip trajectories and unstable posture during grasping, making it difficult to manipulate thin or flat objects under environmental constraints. To overcome these limitations, this paper introduces the DPS (Dual-Parallelogram and Sliding-Guide) hand, a novel robotic gripper that integrates a dual-parallelogram linkage system with a slot-guided sliding mechanism. The combination of the dual-parallelogram structure and the slot-guided mechanism enables the fingertip to achieve a stable and highly linear trajectory, supporting precise and consistent grasping motions. By coordinating these mechanisms with passive elastic elements, the DPS Hand achieves linear-parallel pinching, self-adaptive grasping, and posture-preserving environmental interaction without the need for complex control strategies. A complete kinematic model is developed to characterize the fingertip trajectory and analyze the motion transmission conditions in both linear and adaptive phases. A 3D-printed prototype is fabricated for experimental evaluation, and the results demonstrate that the DPS Hand can reliably grasp thin planar objects, irregular shapes, and multi-sized items with high stability and compliance. The proposed DPS architecture offers a compact and mechanically efficient solution for grasping tasks in both structured and unstructured environments.

I. INTRODUCTION

Robotic end-effectors have evolved from rigid industrial clamps to dexterous systems capable of operating in unstructured environments. Early multi-fingered hands such as the Robonaut Hand demonstrated advanced manipulation capabilities through sophisticated multi-joint actuation [1]. Similarly, systems like the DLR/HIT Hand achieved versatile grasping via multisensory feedback and multi-DOF joint structures [2]. Although these dexterous hands provide high performance, their reliance on independently actuated joints results in large size, high cost, and complex control requirements [3]. The Stanford/JPL Hand further highlighted these limitations by demonstrating that highly

articulated systems often suffer from excessive mechanical and computational complexity [4]. These challenges have stimulated a shift toward underactuated hands, which exploit mechanical intelligence to reduce actuation complexity while maintaining functional adaptability [5]. Foundational studies such as Birglen and Gosselin’s kinetostatic analysis established theoretical frameworks for efficient underactuated design [6]. Later, the mechanism topology research by Feng et al. demonstrated how structured kinematic units can be reorganized into powerful adaptive designs [7]. Mechanical solutions such as the Gosselin underactuated finger with return actuation showed that passive elements can improve stability without increasing motor count [8]. Practical systems like the SDM Hand also validated that compliant underactuation can achieve robust grasping with simplified construction [9]. A key limitation of traditional parallel grippers is their inherent curved fingertip trajectories, which hinder the grasping of thin or planar objects in real environments [10]. To address this, recent research has introduced geometric straight-line mechanisms for fingertip motion. The IROS 2024 geometrical-structure hand demonstrated that proper linkage coordination can generate precise linear trajectories for parallel pinching [11]. Similarly, double-linkage-slider structures have been shown to support coupled self-adaptive behavior while maintaining controlled fingertip alignment [12]. The rack-crank-slider mechanism also provided a compact solution for achieving parallel-self-adaptive hybrid grasping [13]. In parallel, environment-driven grasping has gained increasing attention, particularly for tasks requiring scooping or lifting thin objects under constrained contact conditions. Yoon and Kim proposed a fully passive finger capable of exploiting environmental constraints for human-inspired adaptive grasping [14]. Scooping-oriented designs have demonstrated strong advantages in credit-card picking, edge lifting, and surface-constrained manipulation. For example, the prying-grasp strategy developed by Zhang et al. enables reliable picking of thin objects using thick fingertips by leveraging passive environmental interactions [15]. Motion-controlled scooping with two-fingered grippers has further expanded the operational envelope for bin-picking and cluttered-scene manipulation [16]. Although existing mechanisms achieve either high-precision linear pinching or environment-assisted scooping, very few designs integrate both capabilities into a single, compact, underactuated structure. Solutions relying on complex straight-line linkages often increase manufacturing difficulty and introduce alignment sensitivity. On the other hand, scooping-focused mechanisms

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typically lack strictly linear fingertip trajectories, reducing their suitability for tasks requiring precise parallel alignment. To bridge this gap, this paper proposes the DPS Hand, a novel gripper that combines a dual-parallelogram linkage with a slot-guided sliding mechanism to achieve stable straight-line fingertip motion, self-adaptive grasping, and environmental scooping within a unified architecture. The dual-parallelogram structure maintains a fixed distal phalanx orientation, while the slot-guided constraint ensures a highly linear and stable fingertip trajectory—significantly improving the gripper’s ability to handle thin planar objects. Moreover, passive elastic components and linkage coordination enable seamless switching between linear pinching, Adaptive Grasping and Environmental Scooping, thus extending the capabilities demonstrated in earlier multimodal adaptive grippers. By integrating precision kinematics with environment-driven functionality, the DPS Hand addresses core limitations of existing underactuated designs and offers a compact, manufacturable, and highly adaptable solution for industrial automation, service robotics, and constrained-environment manipulation.



Fig. 1. The DPS Hand.

The remainder of this paper is organized as follows. Section II introduces the design of the gripper. Section III presents the kinematic analysis and parameter optimization. Section IV evaluates the performance through experiments. Section V concludes the paper and discusses future research and development directions.

II. DESIGN OF THE DPS HAND

This section provides a detailed explanation of how the mechanical architecture of the DPS Hand is designed to achieve its intended functional capabilities, with each core feature discussed in sequence.

A. Design for Environmental Adaptability

In the mechanical design of the DPS Hand, the vertically oriented linear mechanism is crucial for achieving stable thin-object picking. By strictly constraining the fingertip trajectory to the vertical direction, the DPS Hand can generate natural buffering when contacting the tabletop or the

edge of a card, reducing undesirable interactions such as pushing, sliding, or misalignment that typically occur with curved fingertip trajectories. This linear motion not only enhances the hand’s adaptability to tabletop environments but also provides a stable pose foundation for subsequent actions such as surface attachment, insertion, and lifting of thin objects. When designing the linear mechanism, several factors must be considered, including the linearity of the trajectory, the effective workspace, structural compactness, and the horizontal offset relative to the base. Trajectory linearity determines the precision of thin-object contact, the workspace range affects adaptability to different thicknesses and placement conditions, and structural simplicity together with an appropriate offset ensures that the overall layout avoids interference and unnecessary complexity. Based on these requirements, this work designs a vertical linear-motion structure for the DPS Hand that is well suited for thin-object picking tasks. As shown in Fig. 2, the DPS Hand proposed in this study adopts a combined structure consisting of a slot-guided mechanism and dual-parallelogram linkages. Through appropriate linkage ratio configuration and limiter design, the fingertip is able to achieve strictly vertical linear displacement. This architecture preserves the straight-line motion characteristics while significantly simplifying the overall layout, thereby improving structural compactness and manufacturability. In addition, the arrangement effectively prevents interference between the left and right fingers during horizontal pinching, thus expanding the range of object sizes that can be grasped. The kinematic characteristics and simulation validation of this linear mechanism will be discussed in detail in subsequent sections.

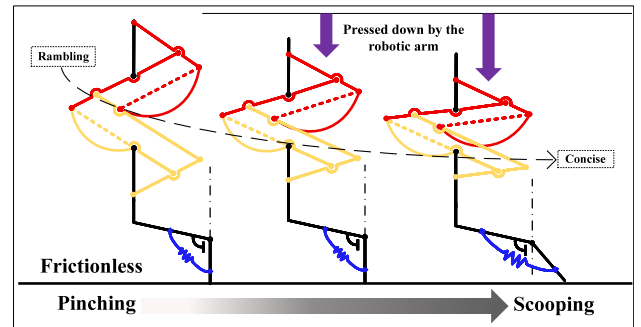


Fig. 2. Deformation process of the mechanism under applied pressure.

B. Design of Adaptive Pinching

The dual-parallelogram–slot composite mechanism adopted in the DPS Hand enables the fingertip to generate a highly linear motion trajectory during the pinching process, ensuring that the distal end maintains a stable orientation before and after contacting the object. Compared with traditional curved fingertip trajectories, a linear path effectively prevents concentrated early-contact points from producing overturning torque, while also improving force distribution and enhancing grasping stability. As shown in Fig. 1, the mechanism exhibits strictly vertical motion as it approaches the object, allowing the left and right fingers to perform symmetric pinching with consistent orientation.

To achieve posture constraint while preserving linear motion, the DPS Hand incorporates paired parallelogram mechanisms. The first-stage parallelogram constrains the inclination of the intermediate link, maintaining a fixed orientation throughout the entire range of motion. The second-stage parallelogram further propagates this orientation relationship to the terminal execution unit, ensuring that the fingertip remains approximately vertical during the linear pinching phase. For aesthetic and structural stability considerations, two of the links in the parallelogram mechanisms are designed as curved members; however, in the subsequent analysis, they are treated as straight links. The combination of the dual-parallelogram mechanism with the slot-guided structure enables both stages of posture constraints to act reliably on the fingertip, thereby achieving linear closure without rotational deviation. This design ensures that the DPS Hand maintains a non-tilting fingertip posture prior to contacting the object, preventing undesired lateral forces that could destabilize thin or edge-like items. On this basis, as shown in Fig. 3, the distal phalanx of the DPS Hand is equipped with a unidirectional rotational constraint, forming the core of its adaptive mechanism. Specifically, for the left finger, the distal link is allowed to rotate freely in the counterclockwise direction, while clockwise rotation is blocked by a stopper, creating a single rotational degree of freedom. Meanwhile, a tension spring connecting the fingertip plate and the curved distal link provides a restoring torque in the initial state, keeping the fingertip plate vertically aligned before contact. When the linear pinching motion encounters irregular shapes or local protrusions, the distal phalanx passively yields due to contact reaction forces, achieving compliant surface alignment. Once the fingertip fully conforms to the object, the structure re-enters a constrained state, forming a stable force-transmission pathway for grasping. Together, the unidirectional yielding and restoring constraints form the basis for transitioning the DPS Hand from standard linear pinching to adaptive grasping.

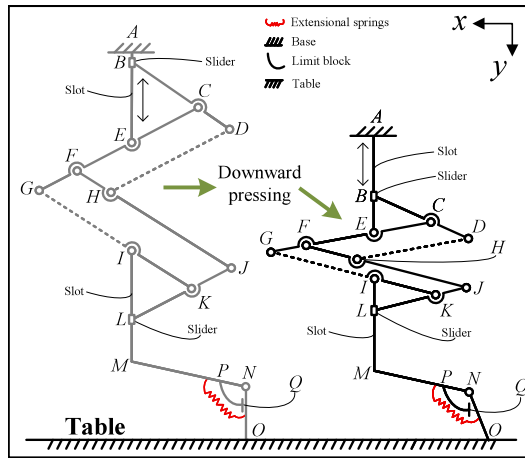


Fig. 3. Structure of the Slot-Guided Mechanism and Dual-Parallelogram Linkages.

This allows the hand to achieve high adaptability and high

stability across diverse scenarios while maintaining structural simplicity.

C. Automatic Switching Between Adaptive Pinching and Card Scooping

The DPS Hand achieves automatic switching between adaptive pinching and scooping modes through the parallelogram linkages and a stopper PQ mounted on the $I-M-N$ linkage. When the DPS Hand presses downward against the table surface, the angle between MN and the distal phalanx NO gradually increases. As the downward motion continues and this angle further enlarges, the tension spring is stretched, generating a clockwise torque on the fingertip plate NO . This causes NO to tilt, shifting the grasping mode from adaptive pinching to scooping, enabling the gripper to scoop thin planar objects. Once the downward force is released, the fingertip plate NO rotates clockwise under the restoring force of the spring until it contacts the stopper PQ , returning the mechanism to the adaptive pinching mode. During tilted downward pressing, the dual-parallelogram mechanism deforms to assist the scooping motion. As shown in Fig. 3, the B -end of link BD moves vertically downward along slot BE . Meanwhile, the angle between BD and the horizontal plane decreases, driving the first-stage parallelogram $C-D-H-F$ to deform, which further reduces the angle between link GC and the horizontal. Link GC then drives the second-stage parallelogram $G-F-J-K$ to deform, causing the angle between link JL and the horizontal plane to decrease; under the constraint of slot IM , the L -end of JL moves vertically upward. At this moment, the entire slot-dual-parallelogram mechanism undergoes compressive deformation, shortening the overall linkage length. Combined with the rotation of the fingertip plate, this deformation enables the automatic transition from adaptive pinching to scooping. The left and right finger mechanisms operate independently. When the DPS Hand remains level, the two fingers perform symmetric pinching and can jointly grasp larger objects. When the DPS Hand is tilted relative to the grasping plane, the finger postures become asymmetric, enabling an asymmetric scooping action that is particularly effective for lifting thin sheet-like objects.

III. THEORETICAL ANALYSIS

A. Kinematic Analysis

To quantitatively characterize the motion of the proposed mechanism, a global Cartesian coordinate system $O-xy$ is established. The origin O is located at the base of the vertical slot AE , with the y -axis aligned with the slot ($x_B = 0$). The mechanism maintains its terminal orientation and linear trajectory through the synergistic coupling of dual-parallelogram units and prismatic constraints. **Cascaded Parallelogram Constraint Derivation:** The structural alignment is preserved by cascading two parallelogram units: $FCDH$ and $GFJK$. First, for unit $FCDH$, the closed-loop geometric constraint is defined by:

$$\vec{r}_{FC} + \vec{r}_{CD} = \vec{r}_{FH} + \vec{r}_{HD}. \quad (1)$$

Given the rigid-link design where $L_{FC} = L_{DH}$ and $L_{CD} = L_{FH}$, it follows that $\vec{r}_{FC} \parallel \vec{r}_{HD}$. Second, for the subsequent unit $GFJK$, the loop equation is:

$$\vec{r}_{GF} + \vec{r}_{FJ} = \vec{r}_{GK} + \vec{r}_{KJ}. \quad (2)$$

With $L_{GF} = L_{KJ}$ and $L_{FJ} = L_{GK}$, we obtain $\vec{r}_{GF} \parallel \vec{r}_{KJ}$. Since the base slot AE provides a fixed vertical reference (orientation $\phi_{\text{base}} = \pi/2$), this orientation is transmitted via the first unit to the intermediate link FC , and subsequently via the second unit to the output rigid link IMN . Consequently, the orientation of the end-effector is locked:

$$\phi_{IMN} = \phi_{\text{base}} = \frac{\pi}{2}. \quad (3)$$

Equation (3) demonstrates that the cascaded parallelogram structure effectively decouples the orientation of IMN from the rotational motion of the internal links. **End-Effector Position and Trajectory Proof:** Let $P_B = [0, y_B]^T$ be the position of the driving slider B in the $O-xy$ frame. The global position of the fingertip N is expressed as the linear summation of the vectors along the kinematic path:

$$\mathbf{P}_N = \mathbf{P}_B + \mathbf{l}_{BD} + \mathbf{l}_{DF} + \mathbf{l}_{FI} + \mathbf{l}_{IN}. \quad (4)$$

Projecting Equation (4) onto the horizontal x -axis, the coordinate is:

$$x_N = x_B + \sum_{i=1}^n L_i \cos(\theta_i). \quad (5)$$

During the structural adaptation process, slider L moves upward within slot IM , inducing the rotation of internal links to condense the vertical profile. Despite this rotation, the horizontal motion is suppressed by the physical guides. Differentiating Equation (5) with respect to time t yields the horizontal velocity.

$$v_{Nx} = \dot{x}_B + \frac{d}{dt}[W(\theta)] = 0. \quad (6)$$

where $W(\theta)$ represents the instantaneous horizontal width. Since $\dot{x}_B = 0$ and the orientation is constrained by (3), v_{Nx} is identically zero, rigorously proving that the trajectory of point N is a strictly vertical straight line. **Kinematic Decoupling and Stability Discussion:** This derivation reveals a fundamental kinematic decoupling between the internal configurational adaptation and the external task-space trajectory. While the internal linkages undergo rotational reconfiguration to achieve vertical collapse—a process captured by the time-varying width $W(\theta)$ —this transformation remains kinematically orthogonal to the output horizontal displacement. This geometric independence, enforced by the holonomic constraints of the dual-parallelogram and vertical guides, ensures that the fingertip N maintains a strictly vertical path throughout the structural collapse. This characteristic is fundamental to achieving superior environmental adaptability. By eliminating parasitic horizontal errors during height adjustment, the mechanism prevents unintended lateral shifts at the contact interface, which significantly enhances grasping stability during complex interaction tasks.

B. Parameter Optimization

To ensure that the DPS linear mechanism maintains structural compactness and balanced load distribution while providing an effective vertical stroke of approximately 200 mm, the primary vertical link lengths must be appropriately allocated. Let l_{BE} denote the effective stroke of point B within its guiding slot in the upper mechanism and l_{LI} the corresponding effective stroke of point L in the lower mechanism. Together, they determine the total available vertical dimension of the mechanism. Considering assembly clearance and safety margins for full downward motion, contacting, pinching, and lifting, the total required vertical height is set to:

$$l_{BE} + l_{LI} = L_{\text{total}} = 0.23 \text{ m}. \quad (7)$$

Under this constraint, the allocation of the two link lengths is treated as an optimization problem. A large difference between l_{BE} and l_{LI} would lead to asymmetric transmission characteristics in the parallelogram chains and reduced structural compactness. Therefore, the following cost function is selected:

$$J = l_{BE}^2 + l_{LI}^2. \quad (8)$$

which penalizes disproportional link distributions. Introducing a Lagrange multiplier λ , the constrained optimization function becomes:

$$\mathcal{L}(l_{BE}, l_{LI}, \lambda) = l_{BE}^2 + l_{LI}^2 + \lambda(l_{BE} + l_{LI} - L_{\text{total}}). \quad (9)$$

Taking partial derivatives with respect to l_{BE} and l_{LI} and setting them to zero yields:

$$\frac{\partial \mathcal{L}}{\partial l_{BE}} = 2l_{BE} + \lambda = 0. \quad (10)$$

$$\frac{\partial \mathcal{L}}{\partial l_{LI}} = 2l_{LI} + \lambda = 0. \quad (11)$$

Subtracting Eqs. (10) and (11) leads to:

$$l_{BE} = l_{LI}. \quad (12)$$

Substituting Eq. (12) into the total-height constraint gives the optimal solution:

$$l_{BE} = l_{LI} = \frac{L_{\text{total}}}{2} = \frac{0.23}{2} = 0.115 \text{ m}. \quad (13)$$

Equation (13) indicates that the optimal configuration is achieved when the upper and lower vertical links have equal lengths. This configuration minimizes J , resulting in the most compact geometry and a symmetric, well-balanced load distribution in the dual-parallelogram mechanism. The remaining link lengths in the triangular structures $B-C-E$ and $L-K-I$ are determined by their similarity ratios. Let:

$$c_1 = \frac{l_{BC}}{l_{BE}} = \frac{l_{LK}}{l_{LI}}, \quad c_2 = \frac{l_{CE}}{l_{BE}} = \frac{l_{KI}}{l_{LI}}. \quad (14)$$

Substituting the optimized value $l_{BE} = l_{LI} = 0.115 \text{ m}$ gives:

$$l_{BC} = c_1 l_{BE}, \quad l_{CE} = c_2 l_{BE}. \quad (15)$$

$$l_{LK} = c_1 l_{LI}, \quad l_{KI} = c_2 l_{LI}. \quad (16)$$

For practically favorable geometry and manufacturability, the ratios $c_1 \approx 1.3$ and $c_2 \approx 0.95$ are selected, leading to the following optimized link set:

$$l_{BE} = l_{LI} = 0.115 \text{ m}. \quad (17)$$

$$l_{BC} = l_{LK} \approx 0.150 \text{ m}. \quad (18)$$

$$l_{CE} = l_{KI} \approx 0.109 \text{ m}. \quad (19)$$

This optimized link-length configuration satisfies the 200 mm vertical stroke requirement and provides a geometrically compact, symmetric, and mechanically stable structure. The resulting design enhances the linear motion accuracy of the DPS hand and contributes to the stability of both the linear pinching and environmental scooping modes. Fig. 4 shows the MATLAB-simulated diagram of the vertical link length optimization for the DPS Hand.

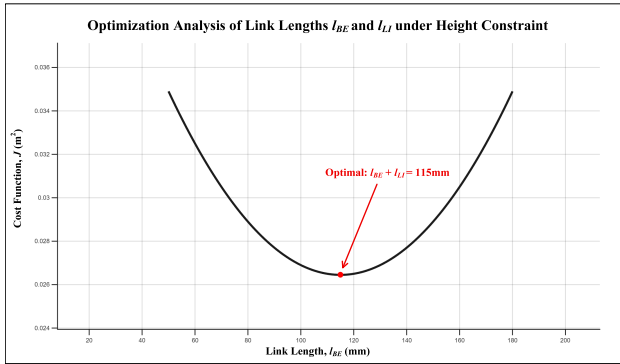


Fig. 4. Optimization of the stroke allocation for the DPS linear mechanism.

IV. EXPERIMENTAL VALIDATION

To validate the effectiveness of the proposed mechanism, a functional prototype of the DPS Hand was developed. The main structural components were fabricated primarily using PLA-based 3D printing, while iron shafts were employed at the joints to secure the slot-guided transmission elements and linkage connections. To enhance grasping stability, a 15 mm-thick silicone layer was applied to the fingertip plate. After assembly and geometric calibration, the prototype underwent a comprehensive series of experiments to evaluate its linear pinching behavior, adaptive grasping performance, and environmental scooping capability. The tests cover both geometry-sensitive thin-object picking and compliance-sensitive grasping of irregular objects. Experimental results demonstrate that the DPS Hand can reliably perform both straight-line pinching and passive scooping interactions, confirming the practical feasibility of the proposed design.

A. Card Scooping

The DPS Hand can automatically switch from linear pinching to card scooping when downward pressure is applied to the table. As the distal phalanx encounters resistance, the unidirectional rotational constraint allows passive yielding, while the spring and stopper drive the fingertip plate to tilt and form a scooping posture. This asymmetric interaction increases the effective contact area and helps lift thin planar

objects such as playing cards and paper sheets, as shown in Fig. 5. Compared with direct pinching, this mode converts part of the downward approach into an insertion-and-lifting motion, reducing dependence on precise edge prealignment. The passive transition also avoids additional sensing or explicit mode commands during tabletop operation.

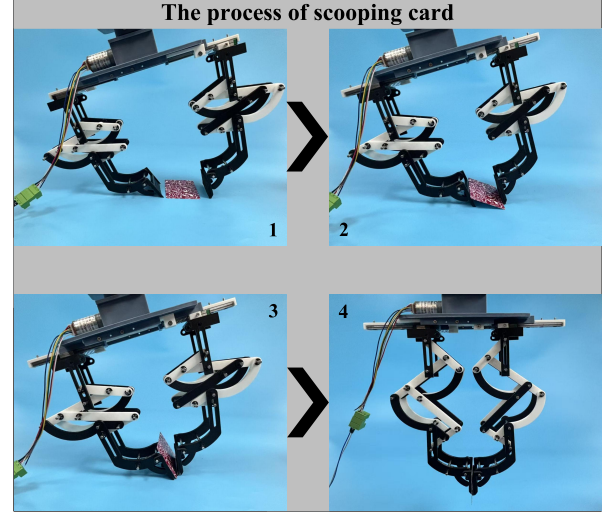


Fig. 5. The card-scooping process.

B. Adaptive Pinching and Lifting

The adaptive pinching experiments were conducted on four representative objects with different geometries: a cubic tea box, a cylindrical plastic bottle, an irregular 3D-printed PLA part, and a slender PLA rod. In all cases, the DPS Hand formed stable grasps, maintained a consistent fingertip posture, and lifted the objects cleanly from the supporting surface. These results demonstrate that the dual-parallelogram and slot-guided mechanism can preserve alignment while adapting to objects of different shapes and sizes, as illustrated in Fig. 6.



Fig. 6. Adaptive pinching and lifting performance test.

C. Grasping Performance and Reliability Evaluation

To evaluate reliability, the prototype was mounted on a CArm robotic arm produced by Guangzhou Shiyuan Electronic Technology Co., Ltd. under the MAXHUB brand and tested on a structured tabletop with a static friction coefficient of approximately 0.4. For each target object category, 100 attempts were conducted under the same initial pose, resulting in 800 trials in total. A trial was counted as successful only if the object was lifted and held stably for 5 s without slipping, thereby excluding transient contacts or momentary lifting events. The lower success rate for the 0.1 mm hard card indicates that the task is mainly limited by initial edge engagement rather than the subsequent holding phase. The DPS Hand achieved an overall success rate of 99.13%, including 100% success for all regular, organic, and irregular objects in pinching and adaptive modes, and 95.00% and 99.00% for 0.1 mm hard cards and 0.25 mm playing cards, respectively, as summarized in Table I. These results suggest that the two operating modes fail for different reasons. Once the object is fully engaged, the pinching mode is highly repeatable, whereas the scooping mode is more sensitive to the initial entrance of the thin edge beneath the fingertip plate. This distinction is useful for future optimization because it points to fingertip entry geometry, stopper timing, and local compliance as the main variables governing thin-object performance.

TABLE I

EXPERIMENTAL RESULTS OF GRASPING PERFORMANCE FOR THE DPS HAND

Object Category	Target Object	Mode	Success Rate
Ultra-thin	Hard Card (0.1 mm)	Scooping	95.00%
Thin sheet	Playing Card (0.25 mm)	Scooping	99.00%
Regular	Hollow Iron Box	Pinching	100.00%
Large cylinder	Plastic Bottle ($\phi 80$ mm)	Pinching	100.00%
Slender	PLA Rod ($\phi 4$ mm)	Pinching	99.00%
Organic sphere	Fresh Orange	Pinching	100.00%
Organic sphere	Fresh Apple	Pinching	100.00%
Irregular	3D Printed Part	Pinching	100.00%
Total / Avg	-	-	99.13%

V. CONCLUSIONS

This paper presented the DPS Hand, an underactuated gripper that combines a slot-guided straight-line mechanism with dual-parallelogram posture constraints to achieve linear pinching and environmental scooping in a unified structure. The proposed mechanism enables stable grasping of thin, irregular, and multi-sized objects while using environmental contact to assist scooping and lifting. Compared with conventional underactuated fingers that mainly rely on curved closing trajectories, the DPS Hand provides a more predictable fingertip path before contact, which is beneficial for thin-object manipulation and tabletop interaction. The slot-guided architecture can also be extended to other straight-line mechanisms, such as Chebyshev, Peaucellier-derived, or Watt-type linkages, according to different compactness and performance requirements. Nevertheless, the current prototype is still limited by manufacturing tolerance, joint clearance, and the stiffness of 3D-printed components, which should be

considered when scaling the mechanism to higher payloads or longer service cycles. Future work will further refine parameter selection, conduct more detailed force-transmission and durability tests, and extend the DPS framework to multi-fingered manipulation. The present experiments also indicate that thin-card retrieval remains the most sensitive task, so future tuning should focus on edge capture and passive compliance rather than only on link-length scaling. Direct comparisons under matched friction and payload conditions will further clarify the performance boundary of the DPS architecture. A Jacobian-based analysis could further guide link-length selection by balancing fingertip speed and force transmission at the workspace. Direct comparisons under matched friction and payload conditions will help isolate the structural benefit of the DPS Hand from the effects of compliance and contact conditions.

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